

An improved and reliable molecular sexing technique for Asiatic black bears, *Ursus thibetanus*

Mukesh · L. K. Sharma · S. A. Charoo ·
S. Sathyakumar

Received: 21 January 2013 / Accepted: 19 June 2013 / Published online: 28 June 2013
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Abstract We developed an improved and reliable molecular genetics technique for sexing Asiatic black bear, *Ursus thibetanus*. Individual sex determination based on differential banding patterns of *SRY* and *Amelogenin* genes were taken advantageously to perform a duplex PCR and sexes were correctly assigned to known and unknown individuals using capillary electrophoresis. Molecular sexing based on fluorescent labeled primers was also found to be appropriate for non-invasive samples, and raised the applicability of multiplexing with microsatellites.

Keywords Molecular sexing · *SRY* · *Amelogenin* · Asiatic black bear · Dachigam National Park

Introduction

Sex determination in natural populations is vital to understand the population dynamics, social organization and making associated management decisions (Eggert et al. 2003; Pilgrim et al. 2005; McEwing et al. 2011). Sex ratio estimation is crucial when managing endangered species or isolated populations as they are believed to be more prone to diversity loss and extinction risk (Frankham 2005). In

many species, sex identification using conventional methods is difficult due to lack of sexual dimorphism, animal behaviour, and limitations in observing rare and elusive species in wild. However, non-invasive genetic sampling has revolutionized wildlife research particularly for effective monitoring and understanding of population dynamics of elusive species that are otherwise difficult to study (Taberlet et al. 1999; Bradley et al. 2001; Waits and Paetkau 2005; Karamanlidis et al. 2010; Thakur et al. 2011).

Although, the Asiatic black bears are widely distributed in the forested habitats of the Himalayas and hills of north east India, they are threatened throughout their distribution range due to various factors viz., habitat degradation and fragmentation, poaching for their body parts and retaliatory killings due to black bear-human conflicts (Charoo et al. 2011). To assess the sex ratio of the wild populations of Asiatic black bears is extremely crucial for research and monitoring programme. The present study was attempted to standardize a molecular sexing technique based on fluorescent labeled primers to correctly assign the sexes of Asiatic black bears dealing with non-invasive samples. This study was carried out as a part of the research project on 'Ecology and Population genetics of Asiatic black bear in Dachigam National Park, Jammu and Kashmir, India' (2007–2012).

Materials and methods

Sample collection and DNA extraction

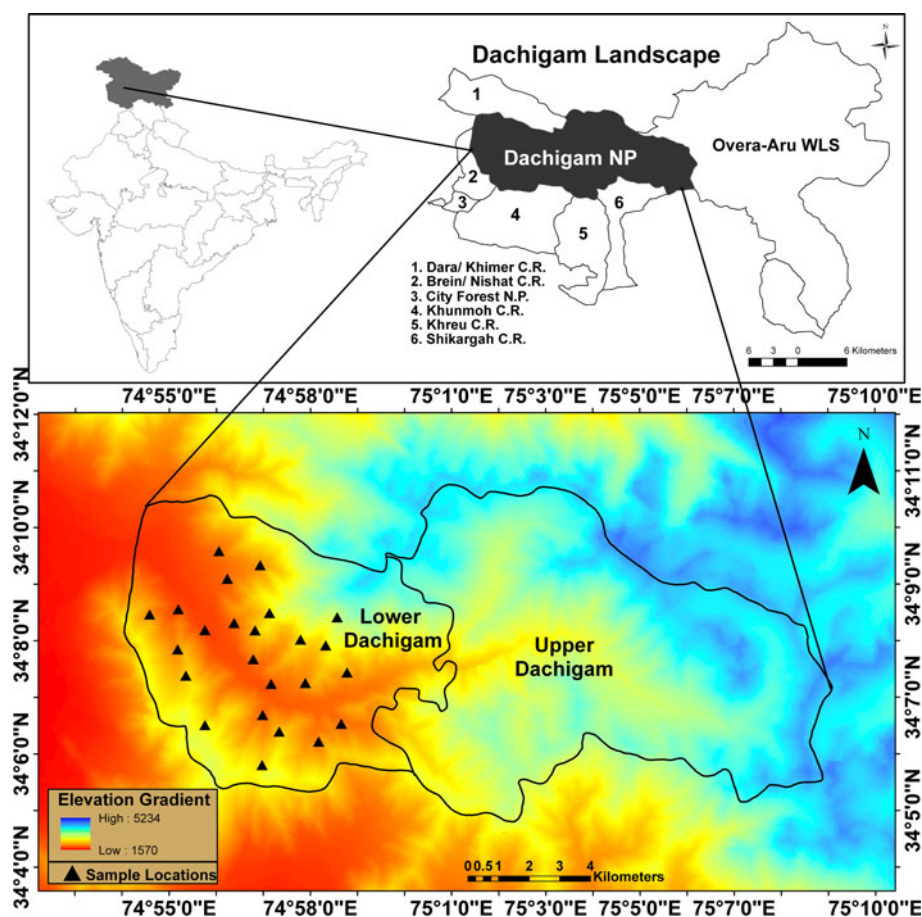
Blood and hair samples were collected from seven radio-collared individuals (four males and three females). In addition, 24 hairs samples of unknown bears were collected from 23 hair snare stations across the study area during March 2011 (Fig. 1). Genomic DNA from the blood spots

Mukesh (✉) · L. K. Sharma · S. A. Charoo ·
S. Sathyakumar (✉)
Wildlife Institute of India, Post Box # 18, Chandrabani,
Dehradun 248001, Uttarakhand, India
e-mail: thamukesh@gmail.com

S. Sathyakumar
e-mail: ssk@wii.gov.in

Mukesh
Department of Biotechnology, Kurukshetra University,
Kurukshetra 136119, Haryana

Fig. 1 Map of Dachigam National Park showing the sampling locations for Asiatic black bear



stored on FTA[®] Classic-Cards (Whatman) was isolated following Smith and Burgoyne (2004) while DNA from hair samples was extracted using DNeasy blood and tissue kit (Qiagen, Germany) following manufacturer's protocol with slight modification (Thakur et al. 2011).

PCR amplification and genotyping

Two genes, *SRY* and *Amelogenin* were selected for molecular sexing of Asiatic black bears. *SRY* gene was amplified following Taberlet et al. (1993) and *Amelogenin* gene was amplified with the primers SE47/SE48 (Ennis and Gallagher 1994; Yamamoto et al. 2002). Since, *SRY* is located on Y-chromosome, a PCR product (ca. 130 bp) is expected to be obtained in males only while *Amelogenin* gene is found on the chromosomes of both sexes and Y-chromosomal copy differs from the X-chromosomal copy by a 52 bp deletion. Thus, amplifications with the SE primers result either in one band (female having two X-chromosomal copies with the expected PCR fragment is of 239 bp) or in two bands (male having X and Y chromosomal copies of different sizes i.e. 239 and 187 bp).

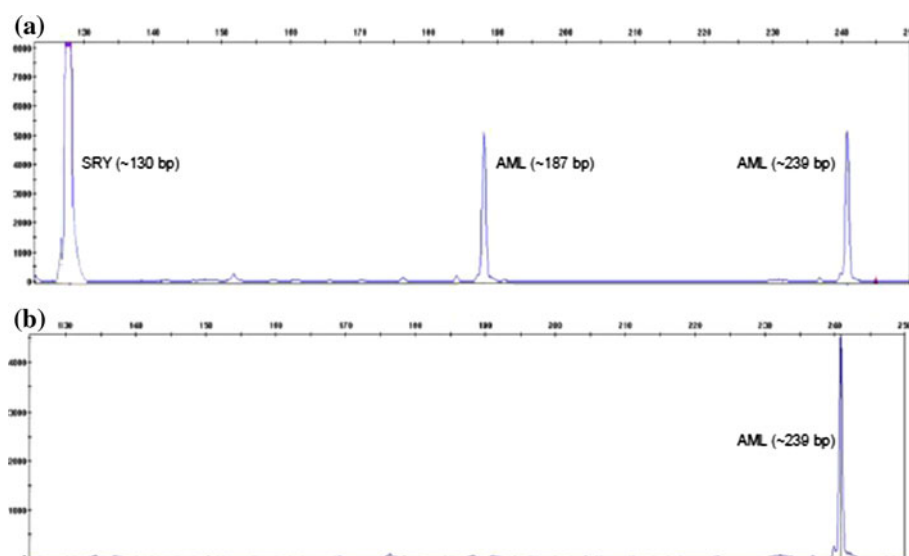
Following Qiagen Multiplex PCR kit (Qiagen, Germany), PCR was set up in a 15 µl reaction containing

2× Qiagen Multiplex PCR Master mix, 10 µM of each primer (including 6-FAM labeled forward and non-labeled reverse primer of *SRY* and *Amelogenin* genes) with approximately 20–30 ng of genomic DNA. The PCR amplification condition included: 94 °C/15 min (Hot Start *Taq* DNA polymerase), 35 × (94 °C/30 s, 60 °C/90 s, 72 °C/60 s) and 60 °C/15 min. Fluorescent based fragment analysis was performed on ABI 3130 Genetic Analyzer (Applied Biosystems, USA) and allele scoring was done using GeneMapper software version 3.7 (Applied Biosystems, USA).

Results and discussion

Initially, sexes of known individuals were assigned using triplicate independent PCRs of *SRY* and *Amelogenin* genes. Thereafter, the primers of both the genes were pooled and sexes were correctly re-assigned using a duplex PCR. As expected, *SRY* and *Amelogenin* genes produced differential allelic patterns in males and females on capillary electrophoresis (Fig. 2). We did not observe any ambiguous genotype on triplicate genotyping of known individuals. However, out of 24 individuals of unknown sexes, *SRY* gene

Fig. 2 Molecular sexing using capillary electrophoresis, *a* electropherogram of a male individual, *b* electropherogram of a female individual



amplification could not be scored confidently in three individuals on agarose gel electrophoresis. But, the fluorescent signal of *SRY* gene was captured for these individuals on capillary electrophoresis. *Amelogenin* gene amplification also identified these bears as male individuals. Thus, all known and unknown individuals were correctly assigned to their respective sexes i.e. 4 males and 3 females in known individuals while 14 males and 10 females in unknown individuals (data of individual identification was not shown). Molecular sexing based on capillary electrophoresis made the experiment more reliable because even a small signal of fluorescence can be detected by capillary electrophoresis. Duplex PCR generated in this study reduced the chances of identification for false female positives that are often misidentified on agarose gel due to low copy number or degraded DNA where Y chromosome markers fail to amplify or the insufficient copies are formed to be scored on gel. Although, other primers (Taberlet et al. 1997; Woods et al. 1999; Kristensen et al. 2011) have also been used for molecular sexing of other bear species, the present study provides an improved and efficient molecular sexing technique for non-invasive samples of Asiatic black bears having the utility of fluorescent labeled primers for *SRY* and *Amelogenin* genes where successful multiplexing with other microsatellite loci was also achieved (Mukesh et al. unpublished data). The effectiveness of the present technique in terms of time, efforts and funds spent for molecular sexing using capillary electrophoresis is being well justified, and will be of use to other researchers working on the population genetics of bear species.

Acknowledgments Authors are thankful to Director, Dean and Research Coordinator, Wildlife Institute of India, Dehradun for their encouragement and support. This study was an outcome of the Research Grant provided by the International Association for Bear Research and Management (IBA). The Department of Wildlife

Protection, Jammu and Kashmir is thanked for granting necessary permission and support during the field work. Mukesh acknowledges the support received from colleagues in the lab.

Conflict of interest None.

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